



Contemporary Experience of Posterior Retroperitoneoscopic Adrenalectomy in the US

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- BACKGROUND:** As the incidence of adrenalectomy increases steadily, so does the use of minimally invasive approaches like posterior retroperitoneoscopic adrenalectomy (PRA). To date, the largest studies of PRA have been from abroad, and we sought to provide a contemporary US update on the outcomes after PRA.
- METHODS:** A retrospective chart review was conducted on all PRAs performed at a single tertiary care institution between 2013 and 2020. Patient demographic characteristics, indication for operation, operative details, and postoperative course were abstracted. Outcomes of interest included 30-day mortality, conversion to open or transabdominal approach, postoperative complication, and 30-day readmission.
- RESULTS:** A total of 249 PRAs were performed between 2013 and 2020. The population was 54.2% women and mean (SD) age was 54.1 (14.1) years. Most lesions (60.6%) were left-sided, and the most common diagnosis was nonfunctioning adenoma (39.4%), followed by pheochromocytoma (21.3%) and aldosteronoma (16.6%). Mean (SD) tumor size was 3.2 cm (range 0.5 to 9.4 cm). Median operative length was 110 minutes (range 30 to 319 minutes). Overall, the complication rate was 6.4%. Nine patients (3.6%) had a minor postoperative complication (Clavien-Dindo I to III) and 5 patients (2.0%) had a major postoperative complication (Clavien-Dindo IV to V), including 1 mortality (0.4%). There were 2 conversions of approach (0.8%). The majority of patients (58.2%) were discharged on postoperative day 1, and 92.0% were discharged by postoperative day 3. The 30-day readmission rate was 1.6%.
- CONCLUSIONS:** Current practice demonstrates that PRA is an extremely safe approach, with a complication rate < 7% and mortality rate < 1%. In addition, the vast majority of patients are able to return home in an expedient manner. (J Am Coll Surg 2021;232:815–821. © 2021 by the American College of Surgeons. Published by Elsevier Inc. All rights reserved.)

Adrenalectomy is an important component in the management of adrenal disease. Although overall it remains a relatively rare operation, its incidence has been rising.¹ This is due in part to the rise in incidentally found adrenal

adenomas from wide use of cross-sectional imaging, as well as an aging population that is more likely to have adrenal pathology.² Given these factors, the frequency of adrenalectomy is likely to continue to rise in the coming years.

Adrenalectomy was originally performed via an open transabdominal operation, but minimally invasive techniques, first introduced in the 1990s, have increasingly become the preferred technique.³ The most common laparoscopic approach has been a transabdominal method, which provides more traditional exposure and anatomy for surgeons.⁴ Around the same time, the posterior retroperitoneoscopic adrenalectomy (PRA) was also introduced; however, its use has only recently begun to become more widely adopted.^{5,6}

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Major potential advantages to PRA are in its avoidance of the abdominal cavity (making prior abdominal operation a nonissue), averting the mobilization of intra-abdominal organs, and sparing intra-abdominal insufflation and its potential physiologic effects.³ Studies from around the world have found PRA to be safe and effective.⁵⁻⁷ To date, many of the largest reviews of this technique have been from Europe.⁷ The aim of our study was to provide a contemporary experience of PRA performed in the US.

METHODS

Data

A retrospective chart review was conducted of all PRAs performed between January 2013 and October 2020 by the Section of Endocrine Surgery at Yale-New Haven Hospital, an academic tertiary care center. PRAs were performed by 4 surgeons during the time period. The study was approved by the Yale Human Investigation Committee.

Variables

Charts were reviewed for patient demographic characteristics, including age, sex, race and ethnicity, BMI, American Society of Anesthesiologists (ASA) physical status classification, and history of abdominal or back operation. In addition, preoperative tumor pathology and size were abstracted. Size was determined as the largest measured diameter by preoperative imaging. Intraoperative data abstracted included length of operation and conversion to transabdominal or open operation. Finally, postoperative information recorded included complications, length of stay, and readmission. Clavien-Dindo classification was used to assess complications as minor (Clavien-Dindo I to III) or major (Clavien-Dindo IV to V).⁸

Surgical procedure

Briefly, the approach that all surgeons in our review took starts with the patient placed in a prone, jack-knifed position. An incision is made just lateral to the 12th rib on the indicated side and a space bluntly created into the retroperitoneum. A 12-mm port is placed to the incision and 2 additional 5-mm ports are placed on either side. Insufflation is created at 20 to 30 mmHg. Dissection is typically performed using a Ligasure device (Medtronic). The paraspinal muscle is identified and followed to the superior pole of the kidney. The plane of the superior pole is dissected and the adrenal gland and vein are isolated. The vein is clipped and divided. The adrenal gland is then dissected free and removed using an Endo Catch bag (Medtronic). Once the gland is removed, the fascia by the 12-mm port is closed, followed by closure

of the skin. Additional specifics of the operative techniques for PRA have been described in detail previously.⁹

Patient selection

Surgical approach was based on surgeon preference and opinion. Overall, there are few contraindications to PRA. The American Association of Endocrine Surgery guidelines recommend open resection for patients with a preoperative diagnosis of adrenocortical carcinoma to better ensure a complete oncologic resection.¹⁰ Some relative contraindications to PRA at our institution include patient inability to tolerate high-pressure insufflation; unfavorable body habitus, particularly excess fat distribution in the back or flank areas; and renal hypertrophy (eg polycystic kidney disease) that limits the potential space between the kidney and adrenal gland required for adequate dissection.

Statistical considerations

Means, medians, and percentages were presented for descriptive analysis. A multivariate logistic regression was created to determine factors associated with developing any complication (eg intraoperative, postoperative, readmission, or conversion). Factors included in analysis were patient age, sex, and BMI; and tumor type, size, and laterality. Age was dichotomized into older than 65 years and younger than 65 years for analysis. Size was categorized into quartiles and those in the top quartile (> 4.1 cm) were classified as “large” tumors and compared with all other tumors. Factors were picked using a combination of backward stepwise regression with a cutoff of $p < 0.2$ for relevance on univariate analysis, along with inclusion of nonsignificant variables thought to hold importance.

RESULTS

Demographics

A total of 249 PRAs were performed during this period, which represented 82% of the 305 adrenalectomies performed by the section during this time. The patient population were predominately female (54.2%) and White (73.5%) patients; mean age was 54.1 years (Table 1). The majority of the population were obese (49.4%) or overweight (31.3%). Roughly one-half had a history of abdominal operation (48.6%). In addition, 16.5% of patients were ASA physical status class II, 81.1% were ASA physical status class III (81.1%), and 2.4% ASA physical status class IV.

The majority of tumors were left-sided (60.6%); 37.8% were right-sided and 1.6% were bilateral. The most common pathologies were benign adenoma (39.4%), pheochromocytoma (21.3%), and aldosteronoma (Conn's syndrome) (16.6%). Mean tumor size was 3.2 cm with

Table 1. Patient Demographic and Tumor Characteristics

Characteristic	Data
Total, n	249
Sex, n (%)	
Female	135 (54.2)
Male	114 (45.8)
Age, y, mean (SD)	54.3 (14.0)
Race or ethnicity, n (%)	
White	183 (73.5)
Black	35 (14.1)
Hispanic	23 (9.2)
Asian or other	6 (2.4)
Unknown or refused	2 (0.8)
BMI, n (%)	
Underweight	1 (0.4)
Normal weight	47 (18.9)
Overweight	78 (31.3)
Obese	123 (49.4)
Surgical history, n (%)	
Abdominal operation	121 (48.6)
Back operation	17 (6.8)
American Society of Anesthesiologists physical status classification, n (%)	
II	41 (16.5)
III	202 (81.1)
IV	6 (2.4)
Tumor type, n (%)	
Functional tumors	
Pheochromocytoma	53 (21.3)
Conn's syndrome	40 (16.6)
Cushing	25 (10.0)
Nonfunctional tumors	
Benign adenoma	98 (39.4)
Metastasis	8 (3.2)
Primary adrenocortical carcinoma	2 (0.8)
Other pathology (eg myelolipoma, hemangioma, cyst, paraganglioma, and androgen-secreting tumor)	23 (9.3)
Tumor side, n (%)	
Right	94 (37.8)
Left	151 (60.6)
Bilateral	4 (1.6)
Tumor size, cm, mean (SD)	3.2 (1.7)

a range of 0.5 to 9.4 cm. Seventeen tumors were not measured preoperatively.

Operative and hospital course

Mediantime of operation from incision to closure was 110 minutes (range 30 to 319 minutes). Average operative time decreased from a median of 141 minutes per operation in 2013 to 90 minutes per operation in 2020 (Fig. 1).

In total, 58.2% of patients were discharged on postoperative day 1. An additional 27.3% were discharged on postoperative day 2 and 6.4% on postoperative day 3. Only 2.4% of patients required a stay of 1 week or longer (Fig. 2).

Complications

Overall, 6.4% of patients (n = 16) had an intraoperative complication, postoperative complication, or readmission (Table 2). There were 7 patients with deviation from

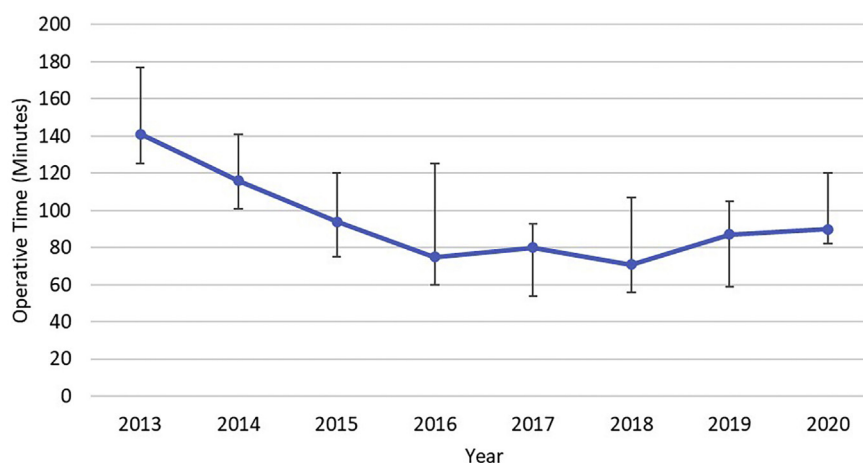


Figure 1. Median operative time (minutes) and range by year. Line graph demonstrating the median operative time by year for the duration of the study.

intraoperative course, including 5 patients who required a blood transfusion. Two patients had a conversion of operative approach to transabdominal, 1 to open and the other to laparoscopic transabdominal, for a rate of 0.8%; both patients had uneventful postoperative courses. In terms of postoperative complications, 9 patients had minor complications (Clavien-Dindo I to III). Three of these patients required a postoperative blood transfusion, 4 had an infection (ie urinary tract infection, pneumonia, or wound infection), 1 had a retroperitoneal hematoma that did not require intervention, and 1 had nausea and vomiting that caused dehydration and readmission. Five patients had major complications (Clavien-Dindo IV to V) and all required ICU admission. One was due to intraoperative hypotension and need for blood pressure support; 1 patient became septic; 1 patient had a pancreatic

injury requiring percutaneous drainage; adrenal insufficiency developed in 1 patient; and the final major complication was a mortality. The 30-day readmission rate was 1.6%, all for minor complications.

The 1 mortality was in a 55-year-old man with a history of poorly differentiated carcinoma of the lung and a 4.5-cm metastasis to his left adrenal. Six months after pneumonectomy and adjuvant pembrolizumab, he underwent an uncomplicated PRA with minimal blood loss. Adrenal insufficiency developed postoperatively and he was started on prednisone. Unfortunately, his course was further complicated by multiorgan failure (respiratory and renal) requiring intubation, pressor support, and dialysis. Given his overall prognosis, his family decided to transition to comfort care and he passed away shortly thereafter. The family declined autopsy.

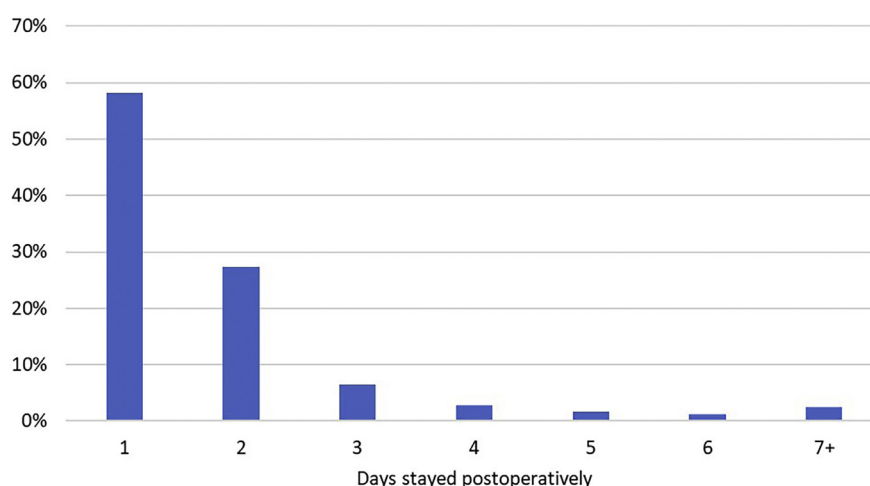


Figure 2. Distribution of length of stay. Bar graph that shows the percentage of patients who were discharged by each postoperative day.

Table 2. Intraoperative and Postoperative Complications

Complication	Data
Intraoperative complication, n (%)	3 (1.2)
Conversion to open, n	2
Intraoperative hypotension, n	1
Postoperative complication, n (%)	14 (5.6)
Minor (Clavien-Dindo I to III), n (%)	9 (3.6)
Blood transfusion only, n	3
Infection (eg urinary tract infection, pneumonia, or wound infection), n	4
Nausea or vomiting, n	1
Retroperitoneal hematoma	1
Major (Clavien-Dindo IV to V), n (%)	5 (2.0)
Surgical ICU admission, n	5
Intrahypotension, n	1
Death, n	1
Pancreatic leak, n	1
Adrenal insufficiency, n	1
Sepsis, n	1
30-d readmission, n (%)	4 (1.6)
30-d mortality, n (%)	1 (0.4)
Any complication, n (%)	16 (6.4)
Total patients, n	249

Odds of complication

Logistic regression demonstrated that the factor most significantly associated with any complication was age older than 65 years (odds ratio 4.48) vs younger than 65 years (95% CI, 1.21 to 16.60) (Table 3). Female patients were also less likely to have complications (odds ratio 0.21) vs male patients (95% CI, 0.05 to 0.90). There was no significant association with complications and race, BMI, type of tumor, or laterality. Notably, there was also no association between large tumors in the top quartile of size (> 4.1 cm) and complication rate.

DISCUSSION

The retroperitoneal approach for adrenalectomy is a safe operation. In our series, the complication rate was 6%, with only 2% of patients experiencing a major complication. Most complications were minor and did not alter recovery significantly. This is in line with, and perhaps even safer than, earlier studies of PRA that demonstrated complication rates ranging from 8% to 17%.^{5,7,11} The largest study of these studies, with 520 patients, from Walz and colleagues,⁷ demonstrated a minor complication rate of 14.4% and a major complication rate of 1.4%. Dickson and colleagues,⁵ in a US case series, found that during a 5-year period, the 47 cases in the first half of

their series had an overall complication rate of 17.2%, and the following 46 had a rate of 7.8%, which is comparable with the rate we experienced in our series. Our large, contemporary series found that modern PRA in the hands of experienced surgeons continues the progression of low complication rates for the retroperitoneal approach.

We were not able to compare with the laparoscopic transabdominal approach directly due to its infrequent use by our surgeons; however, compared with other recent studies of the laparoscopic transabdominal approach, PRA continues to fare well. A recent institutional review of laparoscopic transabdominal adrenalectomy from the University of California, San Francisco, the largest US series to date, demonstrated a complication rate of 8.4%; bleeding was the most common complication.¹² This study, along with our data, showed that both approaches are safe methods with comparable complication rates. There will be great interest in evaluating long-term follow-up from both studies to see whether there is a difference in late complications from these operations. A separate randomized controlled trial comparing laparoscopic transabdominal adrenalectomy with PRA showed a statistically significant increased risk of late complication with the transabdominal approach.¹³ This was almost entirely due to herniation from surgical port sites for transabdominal operations that required additional operative intervention (12.9% compared with 0% for PRA; $p = 0.05$). Additional long-term evaluation in these modern US series is needed to fully compare these approaches.

PRA is an efficient operation with a short length of stay. More than half of patients required only a 1-night stay, and 86% were discharged within 2 days. A European study directly comparing laparoscopic transabdominal adrenalectomy with PRA reported an average length of stay 1.5 days shorter for patients undergoing PRA compared with those undergoing transabdominal adrenalectomy.¹³ This was also a driver of lower overall costs for PRA. Although costs for this European study might not directly translate into the US healthcare system, it shows that the potential for cost savings vs a transabdominal approach exists. Comparison with the previously mentioned US series of laparoscopic transabdominal adrenalectomy also demonstrated slightly shorter lengths of stay. In our study, 14.5% of patients were discharged after postoperative day 2, compared with 15.9% in the transabdominal series. A formal cost analysis is warranted to compare it with the laparoscopic transabdominal approach.

Our experience does highlight a couple of pitfalls for PRA. Although adrenal insufficiency and blood loss are common complications that other series have

Table 3. Factors Associated with Complications

Characteristic	Odds ratio	95% CI
Age		
65 y or older	4.48	1.21–16.60
Younger than 65 y	Ref	Ref
Sex		
Female	0.21	0.05–0.90
Male	Ref	Ref
Race		
White	Ref	Ref
Black	0.85	0.08–8.62
Hispanic	5.82	0.94–35.97
Asian	*	*
Unknown	*	*
BMI		
Underweight	*	*
Normal weight	Ref	Ref
Overweight	0.35	0.08–1.50
Obese	2.30	0.73–7.22
Tumor type		
Adenoma	Ref	Ref
Aldosteronoma	2.71	0.28–25.84
Cortisol secreting	5.36	0.60–48.02
Pheochromocytoma	4.86	0.76–31.01
Other	2.49	0.16–38.97
Tumor size		
Top quartile (> 4.1 cm)	1.87	0.42–8.37
Tumor side		
Left	Ref	Ref
Right	1.24	0.33–4.67
Bilateral	*	*

*Variables with low numbers (< 10) were excluded from analysis.
Ref, reference.

experienced, one unique complication we encountered was pancreatic injury.^{7,9} This has not been a widely reported complication, although it has been seen with transabdominal laparoscopic adrenalectomy.^{14,15} The complication occurred in a male patient with a 1-cm, left-sided tumor, which was managed with pancreatic stent and drainage. Given the proximity of these retroperitoneal organs and difficult orientation, injury to the pancreas is a real concern that surgeons must keep in mind when performing PRA, particularly with left-sided tumors. Unfortunately, there was 1 postoperative death. The exact root cause of this mortality is still unknown, however, contributing factors are thought to be the patient's disseminated cancer and potentially a prolonged autoimmune effect of pembrolizumab, particularly its known potential to facilitate autoimmune hypophysitis.¹⁶

This last point is a good lesson for all general surgeons, as the use of monoclonal antibody therapy continues to rise, surgeons will undoubtedly be increasingly required to operate on patients on these therapies, and they should understand the wide range of impacts it has on surgical patients.

In addition, conversion rates were < 1% in our series. One was due to high end-tidal CO₂ pressures noted intraoperatively. Although the patient had a history of interstitial lung disease that likely contributed, another factor might be the higher CO₂ pressures used in PRA, typically 20 to 30 mmHg compared with a more typical 15 mmHg for transabdominal laparoscopy. Studies have demonstrated that CO₂ insufflation of the retroperitoneum increases PaCO₂ to levels similar to pneumoperitoneum; a higher insufflation pressure creates a theoretical risk of higher PaCO₂.¹⁷ However, this was a rare occurrence, and providers should simply be aware of it, particularly when operating on patients who might have reasons for impaired gas exchange.

Our study has a number of limitations. First, this is a single-institution study that describes the experience of 1 surgical section and results might not apply broadly. In addition, because this is the preferred method of all of our endocrine surgeons, there was not a broad enough database to compare with a transabdominal approach, as those undergoing a transabdominal approach in our database are a biased sample. Another limitation is that between the relatively small number of surgeons and fact that many cases had 2 attendings involved, surgeon volume was unable to be assessed adequately. Surgeon-volume outcomes have been well established and could likely play a role in this specialized operation, and is an area in need of additional study. Finally, long-term follow-up is required to more fully understand the more distant complications of PRA, if any.

CONCLUSIONS

In our series, the largest US series to date, PRA proved to be a safe, effective, and efficient method of adrenalectomy. Given its benefits, PRA is an important approach that should be considered by surgeons for patients with adrenal pathology.

Author Contributions

Study conception and design: Chiu, Vargas-Pinto, Abou-Azar, Maduka, Man, Peters, Carling, Gibson

Acquisition of data: Chiu, Abou-Azar, Maduka, Man, Peters, Carling, Gibson

Analysis and interpretation of data: Chiu, Vargas-Pinto, Carling, Gibson

Drafting of manuscript: Chiu, Carling, Gibson

Critical revision: Chiu, Carling, Gibson

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Invited Commentary



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Posterior retroperitoneoscopic adrenalectomy (PRA) offers numerous benefits compared with traditional laparoscopic operation. It provides direct access to the retroperitoneal space and can avoid intraperitoneal adhesions from earlier operations. In their article “Contemporary Experience of Posterior Retroperitoneoscopic Adrenalectomy in the US,” Chiu and colleagues¹ described a large experience, including 249 cases in 7 years' time. The authors used this technique for a variety of adrenal pathologies, including pheochromocytoma, aldosteronoma, and nonfunctioning adenomas predominantly. They reported a very low rate of complications (7%) and only 1 death in their entire series, which appeared to be unrelated to the operation itself. The retroperitoneoscopic approach to adrenalectomy has become widespread throughout the US and endocrine surgery training programs, predominantly from popularization by Martin Walz of Essen, Germany, who reported on his first series of 27 patients in 1996.² Although Walz has been the driving force in spreading this minimally invasive technique, the posterior approach to adrenalectomy has a much older history.

The first report of a posterior adrenalectomy was from Lennox Ross Broster, a London surgeon, whose 8-year experience with operating on adrenal tumors was published in *Archives of Surgery* in 1937.³ He described the technique briefly as follows: “a sub-diaphragmatic route has now been attempted by fracturing the last rib at its neck and retracting this upward with the diaphragm. The exposure is more difficult, and the delivery of an adrenal gland may be impeded by sudden diaphragmatic movements. . . on the whole, it is a decided improvement, and in our later cases the patient has been discharged within a fortnight of the operation.”

Shortly thereafter, Charles Huggins began using bilateral adrenalectomy combined with orchiectomy for hormonal treatment of metastatic prostate cancers,⁴ as well as bilateral adrenalectomy and oophorectomy for metastatic breast cancers.⁵ This insight led to widespread use of bilateral posterior adrenalectomy for these indications and, in 1966, Huggins won the Nobel Prize for his work.

Although indications for adrenalectomy have changed in the modern era (eg incidentalomas), and the technology has improved dramatically (eg laparoscopy, vessel sealers), the optimal surgical approach remains a matter of debate. The Yale group prefers to perform the operation through the posterior approach, and others have advocated anthropometric measurements to select the approach.^{6,7} As an example, Lindeman and colleagues⁷ created an index called the “Posterior Adiposity Index” to help guide surgeons on the challenges of this approach. One utility of the Lindeman index is that it considers distance from the skin to Gerota's fascia, which can impair trocar mobility when deep or thick. It also accounts for the perinephric fat thickness, which can hinder tumor visualization and dissection in a tight working space. In addition, perirenal fat is often metabolically active and can therefore make the dissection more challenging, especially in chronic states of metabolic dysfunction, such as insulin resistance. This can be seen preoperatively on a CT scan in the visceral and